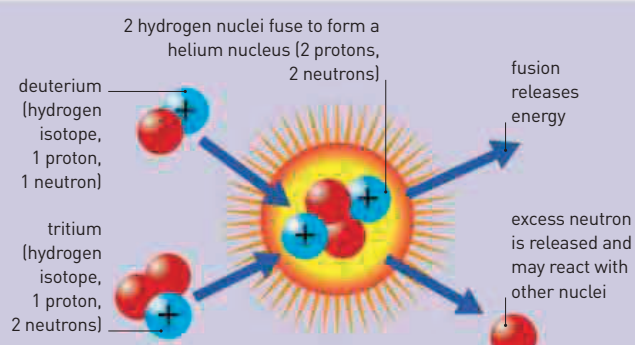


8 THE TYPICAL NUMBER OF CELLS IN AN EMBRYO BEFORE REMOVING A CELL FOR GENETIC DIAGNOSIS

A single cell is removed from an embryo, the first stage of preimplantation genetic diagnosis. The embryo can continue to develop unharmed—the removed cell is subjected to genetic testing.



NUCLEAR FUSION

Fusion is the process of joining together atomic nuclei, most commonly deuterium and tritium (heavy hydrogen) nuclei fusing to form a nucleus of helium. This releases a burst of energy and is the power source deep inside stars. Fusion has been achieved in experiments but, so far, the amount of energy input to create the heat and pressure is much greater than the energy released.

toxin, produced by bacteria in the genus *Clostridium*, can be lethal even in tiny amounts. Upon injection, it causes **paralysis of facial muscles** for around three months. In the same year, American plastic surgeon Richard Clark reported that injections of botox had removed unwanted wrinkling above one eye in a patient who had paralysis in the facial nerve on one side. The fact that botox **injections can reduce wrinkles**—one of the most visible signs of aging—was of great interest to many customers of cosmetic surgeons.

Some began receiving the treatment for cosmetic reasons only, illegally at first; botox injections were approved for cosmetic use in the US in 2002, and shortly afterward in other countries.

In the 11 years since the birth of the first baby conceived by IVF

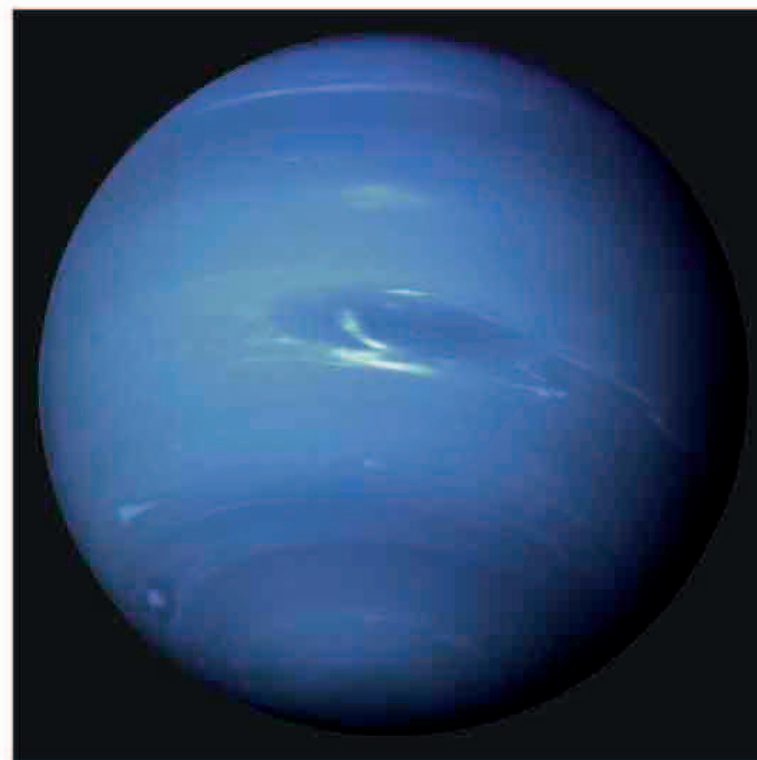
(see 1978), the technology of assisted reproduction advanced greatly. The fact that embryos are created outside the body (and typically implanted after three days' growth) opened up the possibility of **genetic testing on embryos** from parents with certain genetic diseases. Since a cycle of IVF creates several embryos, any that carried the genes giving rise to the disease would be discarded. A team headed by British IVF doctors Alan Handyside (b.1951) and Robert Winston (b.1940) carried out the first human **preimplantation genetic diagnosis** (PGD). The procedure was controversial, with some disability rights groups claiming it was a high-tech version of eugenics.

There was also controversy surrounding a claim made by American physicist **Stanley Pons** (b.1943) and British physicist **Martin Fleischmann** (1927–2012), who were at the University of Utah. The pair announced they had conducted an experiment in which more energy was produced than could be explained by

chemical reactions alone. They claimed that the extra energy had come from nuclear fusion—normally only possible at **extremely high temperatures and pressures** (see panel, left). There was great interest in this “cold fusion,” but many scientists were sceptical, and no one could repeat the experiment with the same result—leading the scientific community to conclude Pons’ and Fleischmann’s **claim was almost certainly incorrect**.

In August, NASA’s Voyager 2 made its closest approach to Neptune, capturing the **first detailed images** of the planet. It was Voyager 2’s last visit to any planet or moon before it headed toward the outer reaches of the Solar System.

Neptune
NASA’s Voyager 2 spacecraft captured images of the gas giant Neptune’s clouds, whose blue color is due to the presence of methane.



12 YEARS THE TIME VOYAGER 2 TOOK TO REACH THE PLANET NEPTUNE

February 14, 1989
The first of a new set (Block II) of GPS satellites launched

March 1, 1989 Discovery of giant magnetoresistance effect, which allows data to be stored at high densities

April 14, 1989 American oncologist Bert Vogelstein discovers the function of p53, the tumor-suppressing protein

August 25, 1989 Space probe Voyager 2’s closest approach to Neptune, capturing stunning images of the planet

March 1989 British computer scientist Tim Berners-Lee writes the blueprint for the World Wide Web

March 23, 1989 Researchers announce that they had achieved cold fusion—a claim later shown to be false

May 11, 1989 Intel announces its 80486 chip, the first to have a million transistors

October 1989 Alan Handyside carries out the first preimplantation genetic diagnosis, screening IVF embryos for genetic diseases before implantation